



Chanakan Arponpong

01

THE CYBERATTACK LIFECYCLE



CSC G0040





ABOUT PROJECT

This project investigates an insider data-leak scenario on a company-issued iPad. The user, who has legitimate access to confidential business files, secretly transfers documents to personal cloud services such as iCloud Drive, Google Drive, or Dropbox. I will simulate file activities—opening, copying, deleting, and uploading—and then examine the digital traces left behind using a full system backup. The backup will be analyzed in Autopsy to identify file interactions, cloud-sync artifacts, and other evidence related to unauthorized data exfiltration.



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METHODOLOGY



**Simulate suspicious actions
on iPad**



**Create iTunes full backup on
Windows**



Transfer backup



Analyze with Autopsy



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OUR TOOLS

01

iTunes



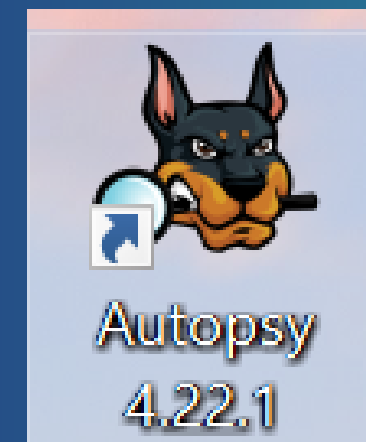
02

Oracle VirtualBox



03

Autopsy





1 iPad Evidence Creation

- Opening, copying, deleting files
- Uploading to Google Drive

2 Backup Acquisition

- Full unencrypted iTunes backup
- Located under MobileSync/Backup
- Transferred into VM for analysis

3 Autopsy Analysis

- Loaded logical evidence
- Examined file system
- Located cloud artifacts & sync logs
- Extracted timeline of user actions

STEPS



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VM (ORACLE)



VM Manager

Help

Open Settings Discard Start

Forensic-VM (Snapshot 3)
Powered Off

Details

General
Name: Forensic-VM
Operating System: Ubuntu (64-bit)

System
Base Memory: 4096 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, KVM Paravirtualization

Display
Video Memory: 136 MB
Screens: 3
Graphics Controller: VMSVGA
3D Acceleration: Enabled
Remote Desktop Server: Disabled
Recording: Disabled

Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: Forensic-VM.vdi (Normal, 25.00 GB)

Audio
Host Driver: Default
Controller: ICH AC97

Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)

Preview
Forensic-





BACK UP

Up

Important data on your iPad to iCloud.

your iPad will be stored on this computer.

Backup

unt passwords, Health, and HomeKit data to be

password...

Manually Back Up and Restore

Manually back up your iPad to this computer backup stored on this computer.

Back Up Now

Restore Backup

Latest Backup:

Today 4:23 PM to this computer



AUTOPSY



4.22.1
File Window Help

Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists

Archives (0)
Databases (3)
Documents
HTML (0)
Office (1)
PDF (0)
Plain Text (0)
Rich Text (0)
Executable
MIME Type
d Files
e System (0)
(0)
ze
acts
Communication Accounts (2)
vice (1)
one (1)
cts (1)
ata (16)
results
tion Suspected (1)
Metadata (45)
Content Suspected (45)
nts
ems (0)
ious Items (1)

Listing

Metadata

Table Thumbnail Summary

Date Created	Organization	Description	Data Source	Program Name
2024-05-30 08:07:24 EDT		Grade	LogicalFileSet1	
2018-05-12 12:08:49 EDT			LogicalFileSet1	
2025-02-06 05:13:43 EST		Copy of Black and White Minimalist Professional Resu...	LogicalFileSet1	
2023-10-18 17:31:37 EDT		Black and White Minimalist Professional Resume	LogicalFileSet1	
		Confidential	LogicalFileSet1	
			LogicalFileSet1	
			LogicalFileSet1	
2025-11-10 23:40:32 EST			LogicalFileSet1	
			LogicalFileSet1	
2024-08-16 12:18:19 EDT		ELI Payment Registration and Payment Form	LogicalFileSet1	
			LogicalFileSet1	
2025-01-27 18:35:04 EST			LogicalFileSet1	

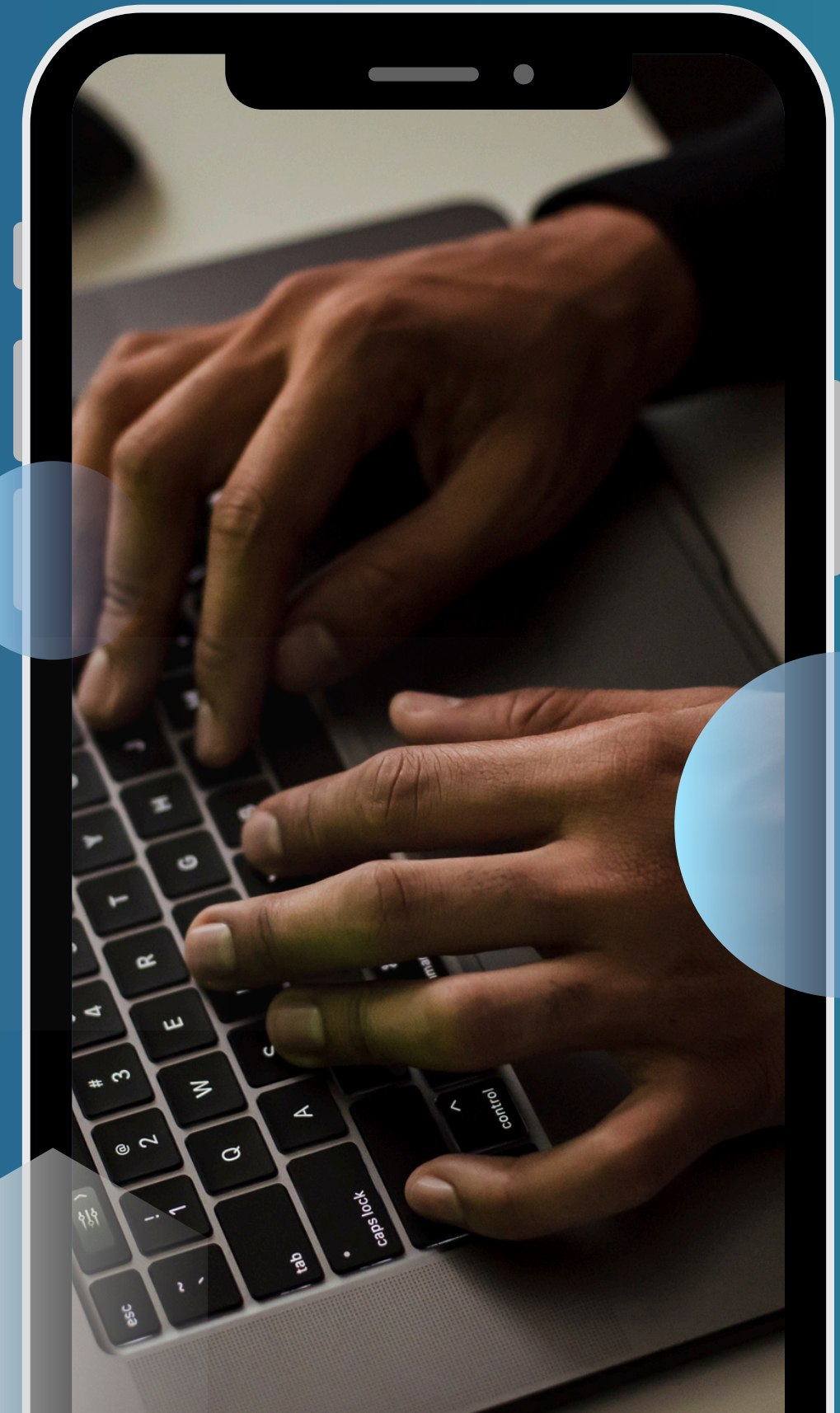
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences





KEY CONCLUSION

- Evidence proves insider exfiltration
- iOS backup preserves strong forensic artifacts
- Autopsy effective for timeline correlation
- Supports internal investigation workflow





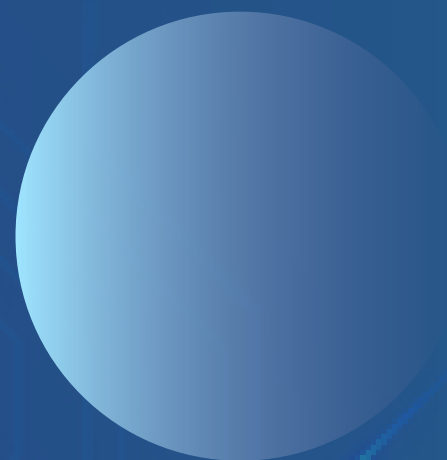
LIMITATIONS

What iTunes Backup does not capture (because Apple does not store or export this data):

- Logs of file opening
- File copy or move actions between folders
- Upload activity to Google Drive / Dropbox / iCloud
- File deletion activity
- Files App usage history
- Detailed event timeline
- Cloud-sync transaction logs

These artifacts are stored in protected system areas that iTunes Backup cannot access or copy.





**THANK
YOU!**

